



Basics of Statistics

5th November, 2023

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What is Statistics

Statistics is a branch of Mathematics which involves collection, analysis and organization of data. Statistics provide various techniques to draw conclusions from the data available.

Different types of Statistics:

Statistics are mainly categorized into two types:

- Descriptive statistics
- Inferential statistics

Data we normally encounter...

POS	Player	Mat	Inns	NO	Runs	HS	Avg	BF	SR	100	50	4s	6s
1	Jos Buttler	17	17	2	863	116	57.53	579	149.05	4	4	83	45
2	K L Rahul	15	15	3	616	103*	51.33	455	135.38	2	4	45	30
3	Quinton De Kock	15	15	1	508	140*	36.29	341	148.97	1	3	47	23
4	Hardik Pandya	15	15	4	487	87*	44.27	371	131.26	0	4	49	12
5	Shubman Gill	16	16	2	483	96	34.5	365	132.32	0	4	51	11
6	David Miller	16	16	9	481	94*	68.71	337	142.72	0	2	32	23
7	Faf Du Plessis	16	16	1	468	96	31.2	367	127.52	0	3	49	13
8	Shikhar Dhawan	14	14	2	460	88*	38.33	375	122.66	0	3	47	12
9	Sanju Samson	17	17	1	458	55	28.63	312	146.79	0	2	43	26
10	Deepak Hooda	15	14	0	451	59	32.21	330	136.66	0	4	36	18
11	Liam Livingstone	14	14	2	437	70	36.42	240	182.08	0	4	29	34
12	David Warner	12	12	3	432	92*	48	287	150.52	0	5	52	15
13	Abhishek Sharma	14	14	0	426	75	30.43	320	133.12	0	2	47	13
14	Ishan Kishan	14	14	1	418	81*	32.15	348	120.11	0	3	45	11
15	Rahul Tripathi	14	14	3	413	76	37.55	261	158.23	0	3	40	20
16	Shreyas Iyer	14	14	1	401	85	30.85	298	134.56	0	3	41	11
17	Tilak Varma	14	14	3	397	61	36.09	303	131.02	0	2	29	16
18	Aiden Markram	14	12	4	381	68*	47.63	274	139.05	0	3	25	19
19	Devdutt Padikkal	17	17	0	376	54	22.12	306	122.87	0	1	40	14
20	Ruturaj Gaikwad	14	14	0	368	99	26.29	291	126.46	0	3	33	14

Different Types of Statistics

Descriptive Statistics:

Descriptive statistics involves organizing and summarizing the data. It involves calculation of central tendency. Central tendency is a statistical measure that represents the whole data.

Central tendency involves the calculation of mean, median and mode.

Measure of dispersion involves calculation of range, standard deviation, variance that indicates spread of the data.

Different Types of Statistics(Contd...)

Inferential Statistics:

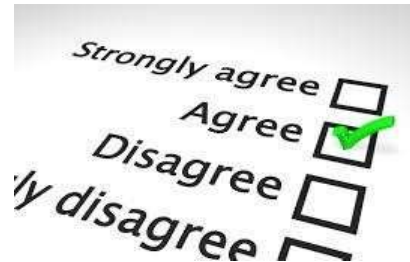
Inferential statistics involves making inferences, drawing conclusions of the population data using the sample data.

It uses sample data to make predictions of population data.

Not all data are of same type...

POS	Player	Mat	Inns	NO	Runs	HS	Avg	BF	SR	100	50	4s	6s
1	Jos Buttler	17	17	2	863	116	57.53	579	149.05	4	4	83	45
2	K L Rahul	15	15	3	616	103*	51.33	455	135.38	2	4	45	30
3	Quinton De Kock	15	15	1	508	140*	36.29	341	148.97	1	3	47	23
4	Hardik Pandya	15	15	4	487	87*	44.27	371	131.26	0	4	49	12
5	Shubman Gill	16	16	2	483	96	34.5	365	132.32	0	4	51	11
6	David Miller	16	16	9	481	94*	68.71	337	142.72	0	2	32	23
7	Faf Du Plessis	16	16	1	468	96	31.2	367	127.52	0	3	49	13
8	Shikhar Dhawan	14	14	2	460	88*	38.33	375	122.66	0	3	47	12
9	Sanju Samson	17	17	1	458	55	28.63	312	146.79	0	2	43	26
10	Deepak Hooda	15	14	0	451	59	32.21	330	136.66	0	4	36	18
11	Liam Livingstone	14	14	2	437	70	36.42	240	182.08	0	4	29	34
12	David Warner	12	12	3	432	92*	48	287	150.52	0	5	52	15
13	Abhishek Sharma	14	14	0	426	75	30.43	320	133.12	0	2	47	13
14	Ishan Kishan	14	14	1	418	81*	32.15	348	120.11	0	3	45	11
15	Rahul Tripathi	14	14	3	413	76	37.55	261	158.23	0	3	40	20
16	Shreyas Iyer	14	14	1	401	85	30.85	298	134.56	0	3	41	11
17	Tilak Varma	14	14	3	397	61	36.09	303	131.02	0	2	29	16
18	Aiden Markram	14	12	4	381	68*	47.63	274	139.05	0	3	25	19
19	Devdutt Padikkal	17	17	0	376	54	22.12	306	122.87	0	1	40	14
20	Ruturaj Gaikwad	14	14	0	368	99	26.29	291	126.46	0	3	33	14

Data Type - Types of data



Description

It is the unique labels or descriptors that are used to designate each value of the scale. All scales possess description



Scale Characteristics

Order

It is the relative sizes or positions of the descriptors. Order is denoted by descriptors such as greater than, less than, and equal to.

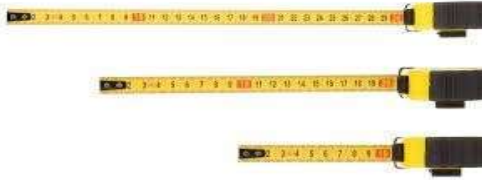


?

Scale Characteristics

Distance

It is the absolute differences between the scale descriptors are known and may be expressed in units.



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Origin

The origin characteristic means that the scale has a unique or fixed beginning or true zero point.



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Types of Data

Nominal

Ordinal



Categorical, Non-Metric, Qualitative

Interval

Ratio



Continuous, Metric, Quantitative

Types of Data

Nominal:

Data is categorized into distinct groups or categories. In this values are assigned to represent different categories, but the values don't have order or numerical meaning.

Example:

Types of cricket shots played by a batsman (e.g., Cover Drive, Pull Shot, Square Cut, etc.)

Types of Data

Ordinal:

Ordinal involves data can be ordered or ranked based on their relative position or preference. It provides a meaningful order among the categories, but the differences between the categories may not be equal or measurable.

Example: ICC annual Ranking of players on the basis of batting performance for the year.

Types of Data

Interval:

The interval represents data with meaningful order and equal intervals between the values. The differences between values are consistent and measurable, but there is no absolute zero point.

Example: Temperature in cricket matches is measured in Celsius or Fahrenheit. The intervals between 20°C and 30°C are equal to the intervals between 30°C and 40°C, but you cannot take any ratios.

Ratio:

The ratio scale has all the characteristics of the interval scale but also has a meaningful and absolute zero point.

Example: Runs scored by cricket players in a match. A player scoring 100 runs has achieved twice as many runs as a player scoring 50 runs. The zero runs (0) represents the absence of runs

Ordinal
Nominal
Interval
Ratio

POS	Player	Mat	Inns	NO	Runs	HS	Avg	BF	SR	100	50	4s	6s
1	Jos Buttler	17	17	2	863	116	57.53	579	149.05	4	4	83	45
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5	Shubman Gill	16	16	2	483	96	34.5	365	132.32	0	4	51	11
6	David Miller	16	16	9	481	94*	68.71	337	142.72	0	2	32	23
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8	Shikhar Dhawan	14	14	2	460	88*	38.33	375	122.66	0	3	47	12
9	Sanju Samson	17	17	1	458	55	28.63	312	146.79	0	2	43	26
10	Deepak Hooda	15	14	0	451	59	32.21	330	136.66	0	4	36	18
11	Liam Livingstone	14	14	2	437	70	36.42	240	182.08	0	4	29	34
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17	Tilak Varma	14	14	3	397	61	36.09	303	131.02	0	2	29	16
18	Aiden Markram	14	12	4	381	68*	47.63	274	139.05	0	3	25	19
19	Devdutt Padikkal	17	17	0	376	54	22.12	306	122.87	0	1	40	14
20	Ruturaj Gaikwad	14	14	0	368	99	26.29	291	126.46	0	3	33	14

Descriptive Statistics

Statistics – Single variable

Three important defining characteristics of any set of data for a given variable are:

- Shape of Data(Also called Distribution)
- Measure of Location(Also called Central Tendency)
- Spread of Data(Also called Variability)

David Warner has played for Sunrisers Hyderabad for many years

Some curious cricket fan wanted to know about David Warner's performance as a batsman. He had a few specific questions answered.

What can we say about his performance as a batsman when he was playing for Sunrisers Hyderabad?

How is it different from his performance as a batsman when he was playing for Dare Devils Delhi?

How is Warner's performance as compared to MS Dhoni's performance as a batsman in IPL?

We are given some **sample data** from the past matches to answer the questions posed

Let's look at the sample data...

David Warner Scores – Sunrisers Hyderabad

Data: (Sample) Runs Scored by
David Warner as a part of Sunrisers
Hyderabad

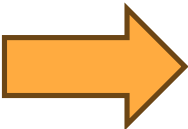
14	8	126
4	44	37
24	0	51
61	61	0
17	13	73
58	4	24
6	28	73
81	57	59
58	53	6
9	6	48
69	40	4
91	52	43
90	46	18
59	30	28
59	11	49
34	92	26
32	52	

What can we say about Warner's performance?

David Warner Scores – Sunrisers Hyderabad

Unordered Data

14	8	126
4	44	37
24	0	51
61	61	0
17	13	73
58	4	24
6	28	73
81	57	59
58	53	6
9	6	48
69	40	4
91	52	43
90	46	18
59	30	28
59	11	49
34	92	26
32	52	



Ordered Data

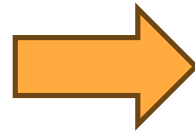
0	26	57
0	28	58
4	28	58
4	30	59
4	32	59
6	34	59
6	37	61
6	40	61
8	43	69
9	44	73
11	46	73
13	48	81
14	49	90
17	51	91
18	52	92
24	52	126
24	53	

Ordering the data provides additional insights quickly

David Warner Scores – Sunrisers Hyderabad

Ordered Data

0	26	57
0	28	58
4	28	58
4	30	59
4	32	59
6	34	59
6	37	61
6	40	61
8	43	69
9	44	73
11	46	73
13	48	81
14	49	90
17	51	91
18	52	92
24	52	126
24	53	

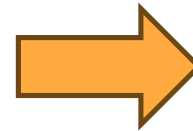


Ungrouped Frequency Distribution

Score Frequency

0	2
1	0
2	0
3	0
4	3
5	0
6	3
7	0
8	1
9	1
10	0
11	1
12	0
13	1
14	1
15	0
16	0
17	1

Up to 126



Grouped Frequency Distribution

Score Frequency

0 - 10	10
11 - 20	5
21 - 30	6
31 - 40	4
41 - 50	5
51 - 60	10
61 - 70	3
71 - 80	2
81 - 90	2
91 - 100	2
101 - 110	0
111 - 120	0
121 - 130	1

OR

Score Frequency

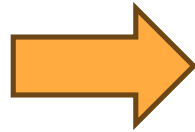
0 - 27	18
28 - 54	16
55 - 81	12
82 - 108	3
109 - 135	1

The right number of groupings and class intervals are subjective and depends on data. Anything over 10 groups becomes difficult to read and comprehend. It is not advisable to vary the class intervals. (e.g. 0 – 40, 41-60, 61 – 90, 91 - 130)

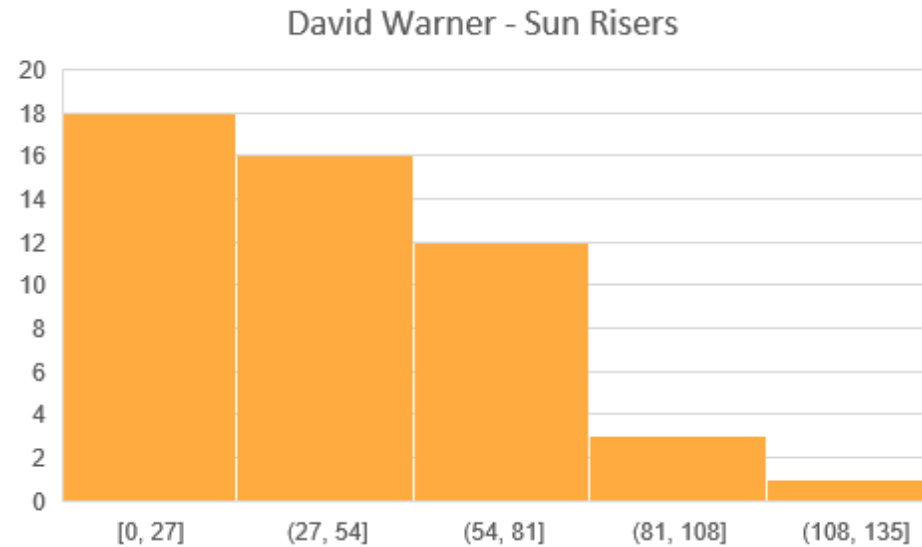
David Warner Scores – Sunrisers Hyderabad

Grouped Frequency Distribution

Score	Frequency
0 - 27	18
28 - 54	16
55 - 81	12
82 - 108	3
109 - 135	1



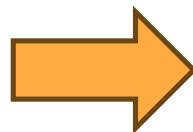
Histogram



David Warner Scores – Sunrisers Hyderabad

Grouped Frequency Distribution

Score	Frequency
0 - 10	10
11 - 20	5
21 - 30	6
31 - 40	4
41 - 50	5
51 - 60	10
61 - 70	3
71 - 80	2
81 - 90	2
91 - 100	2
101 - 110	0
111 - 120	0
121 - 130	1



Relative Frequency Distribution

Score	Frequency	Proportion/Relative Frequency	As a %
0 – 10	10	0.20	20
11 – 20	5	0.10	10
21 – 30	6	0.12	12
31 – 40	4	0.08	8
41 – 50	5	0.10	10
51 – 60	10	0.20	20
61 – 70	3	0.06	6
71 – 80	2	0.04	4
81 – 90	2	0.04	4
91 – 100	2	0.04	4
101 – 110	0	0.00	0
111 – 120	0	0.00	0
121 – 130	1	0.02	2

David Warner Scores – Sunrisers Hyderabad

Relative Frequency Distribution

Score	Frequency	Proportion	As a %
0 – 10	10	0.20	20
11 – 20	5	0.10	10
21 – 30	6	0.12	12
31 – 40	4	0.08	8
41 – 50	5	0.10	10
51 – 60	10	0.20	20
61 – 70	3	0.06	6
71 – 80	2	0.04	4
81 – 90	2	0.04	4
91 – 100	2	0.04	4
101 – 110	0	0.00	0
111 – 120	0	0.00	0
121 – 130	1	0.02	2



Cumulative Relative Frequency Distribution

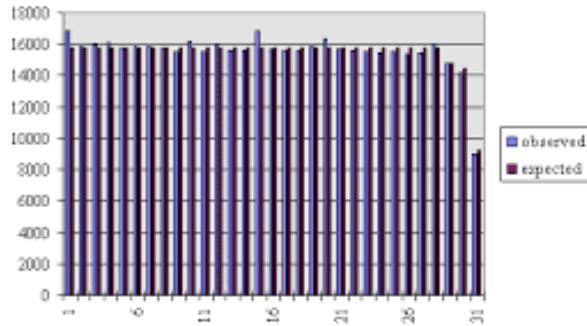
Score	As a %
≥ 0	100
> 10	80
> 20	70
> 30	58
> 40	50
> 50	40
> 60	20
> 70	16
> 80	12
> 90	8
> 100	4
> 110	4
> 120	4

How often does Warner score more than 50?

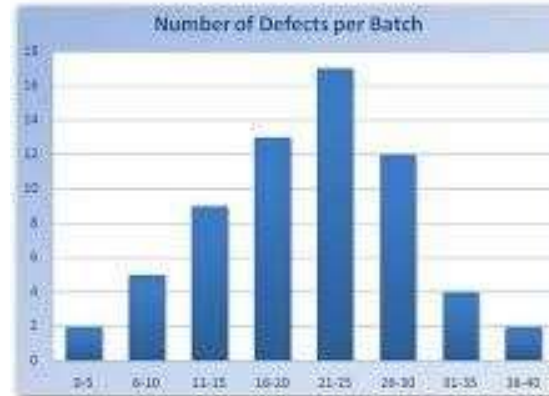
Ans: 40% of Innings

Basic Statistical Measures- Shape of data - Examples

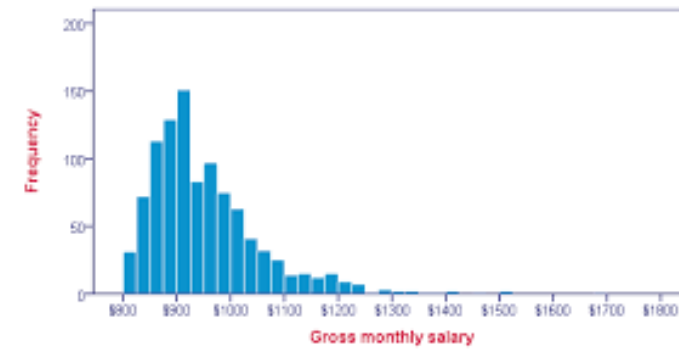
Distribution of Birthdays by Day



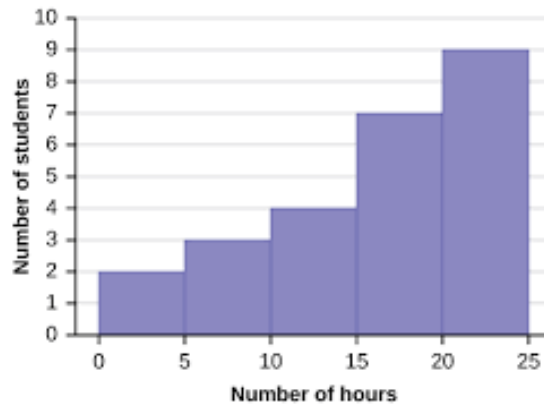
Number of Defects per Batch



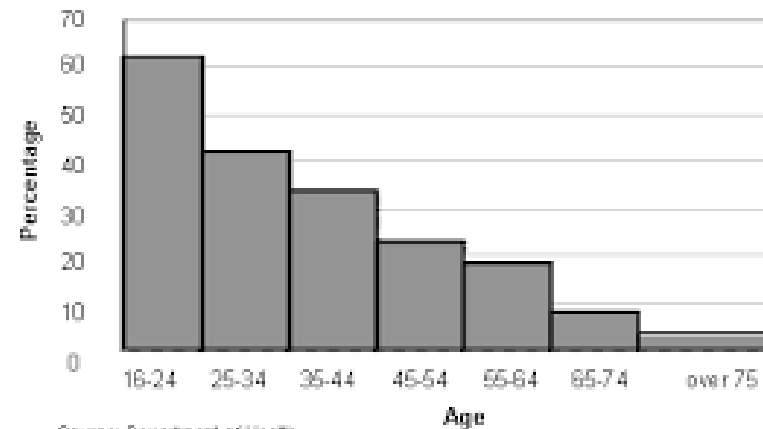
Histogram of Monthly Salary



Hours Spent Playing Video Games on Weekends



Percentage of men spending at least one hour per week participating in sports or exercise, by age



Source: Department of Health

Basic Statistical measures- Measure of location

- The 3 M's, and Proportion
 - Mean
 - Median
 - Mode(and proportion)

The mode is the data value that occurs **most frequently** in the data set

0	26	57
0	28	58
4	28	58
4	30	59
4	32	59
6	34	59
6	37	61
6	40	61
8	43	69
9	44	73
11	46	73
13	48	81
14	49	90
17	51	91
18	52	92
24	52	126
24	53	

Mode: 4 Runs (3/50 – 6%)
6 Runs (3/50 – 6%)
59 Runs (3/50 – 6%)

Advantages:

- Easy to Compute

Disadvantages:

- There may not be a single mode (multimodal data)
- It **does not use all the data** available
- Poor sampling stability - Large variations across samples

David Warner Scores – Mean

The mean is the **arithmetic average** of all the data values in the data set

0	26	57
0	28	58
4	28	58
4	30	59
4	32	59
6	34	59
6	37	61
6	40	61
8	43	69
9	44	73
11	46	73
13	48	81
14	49	90
17	51	91
18	52	92
24	52	126
24	53	

Mean: 40.56 Runs

Advantages:

- Uses all the data values available
- Moderate sampling stability

Disadvantages:

- More **sensitive to extreme values** –Not appropriate for skewed distributions

$$\overline{X} = \frac{\sum X}{N}$$

David Warner Scores – Median

The median is the data value in the distribution that **divides the data into two groups** having equal frequencies – the **center point** of the data set

0	26	57
0	28	58
4	28	58
4	30	59
4	32	59
6	34	59
6	37	61
6	40	61
8	43	69
9	44	73
11	46	73
13	48	81
14	49	90
17	51	91
18	52	92
24	52	126
24	53	

If, n (the number of observations) is odd

Median is the data value of the $((n+1)/2)^{\text{th}}$ Observation, when ordered

If n is even, the Median is the midway point (average) of the $(n/2)^{\text{th}}$ observation, and the $(n/2 + 1)^{\text{th}}$ observation

In this case since we have 50 observations, the median is the average of the 25th, and 26th Observation

Median = 41.5 Runs

In half the innings Warner plays he scores more than 41.5 Runs, and in the other half he scores less than 41.5 Runs

Advantages:

- Simple to compute
- Very appropriate for skewed distributions
- Great sampling stability

Disadvantages:

- Does not use all data values

Summarizing Quantitative Data – Dispersion

Batsman 1: 10, 30, 50, 50, 70, 90 → Mean =50, Median =50, Mode =50

Batsman 2 : 5, 10, 50, 50, 90, 95 → Mean =50, Median =50, Mode =50

Batsman 3 : 50, 50, 50, 50, 50, 50 → Mean =50, Median =50, Mode =50

What is the difference between the three Batsmen?

This difference is characterized as Dispersion (or Variability between the individual data values) and is captured by measures such as **R**ange, **I**nter **Q**uartile **R**ange, **S**tandard **D**eviation

David Warner Scores – Range

The range is defined as the $R = (\text{Maximum Data value}) - (\text{Minimum Data Value})$

0	26	57
0	28	58
4	28	58
4	30	59
4	32	59
6	34	59
6	37	61
6	40	61
8	43	69
9	44	73
11	46	73
13	48	81
14	49	90
17	51	91
18	52	92
24	52	126
24	53	

Range = $126 - 0 = 126$ Runs

Advantages:

- Easy to compute
- Provides a quick understanding of the total spread of the data values

Disadvantages:

- Uses only two data values
- Highly influenced by extreme data values
- Poor sampling stability

David Warner Scores – IQR

0	26	57
0	28	58
4	28	58
4	30	59
4	32	59
6	34	59
6	37	61
6	40	61
8	43	69
9	44	73
11	46	73
13	48	81
14	49	90
17	51	91
18	52	92
24	52	126
24	53	

The **IQR** is the range of the middle % 50 of the data

Some definitions:

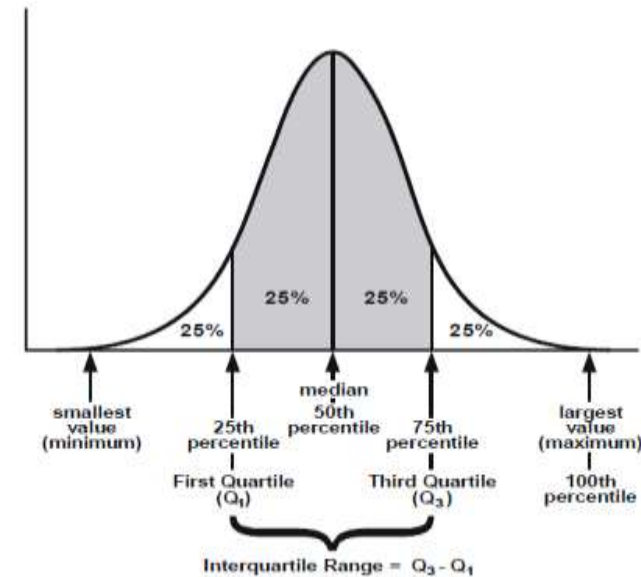
Maximum (100th percentile) = Q_4

75th percentile = Q_3

Median (50th percentile) = Q_2

25th percentile = Q_1

Minimum (0th percentile) = Q_0



Range = $Q_4 - Q_0$ = (100th Percentile – 0th Percentile)

Inter Quartile Range = $Q_3 - Q_1$ = (75th Percentile – 25th Percentile)

David Warner Scores – IQR

0	26	57
0	28	58
4	28	58
4	30	59
4	32	59
6	34	59
6	37	61
6	40	61
8	43	69
9	44	73
11	46	73
13	48	81
14	49	90
17	51	91
18	52	92
24	52	126
24	53	

The **IQR** is the range of the middle % 50 of the data

Some definitions:

Maximum (100th percentile) = Q_4

126

75th percentile = Q_3

59

Median (50th percentile) = Q_2

41.5

25th percentile = Q_1

14

Minimum (0th percentile) = Q_0

0

Range = $Q_4 - Q_0$

126 - 0 = 126 Runs

Inter Quartile Range = $Q_3 - Q_1$

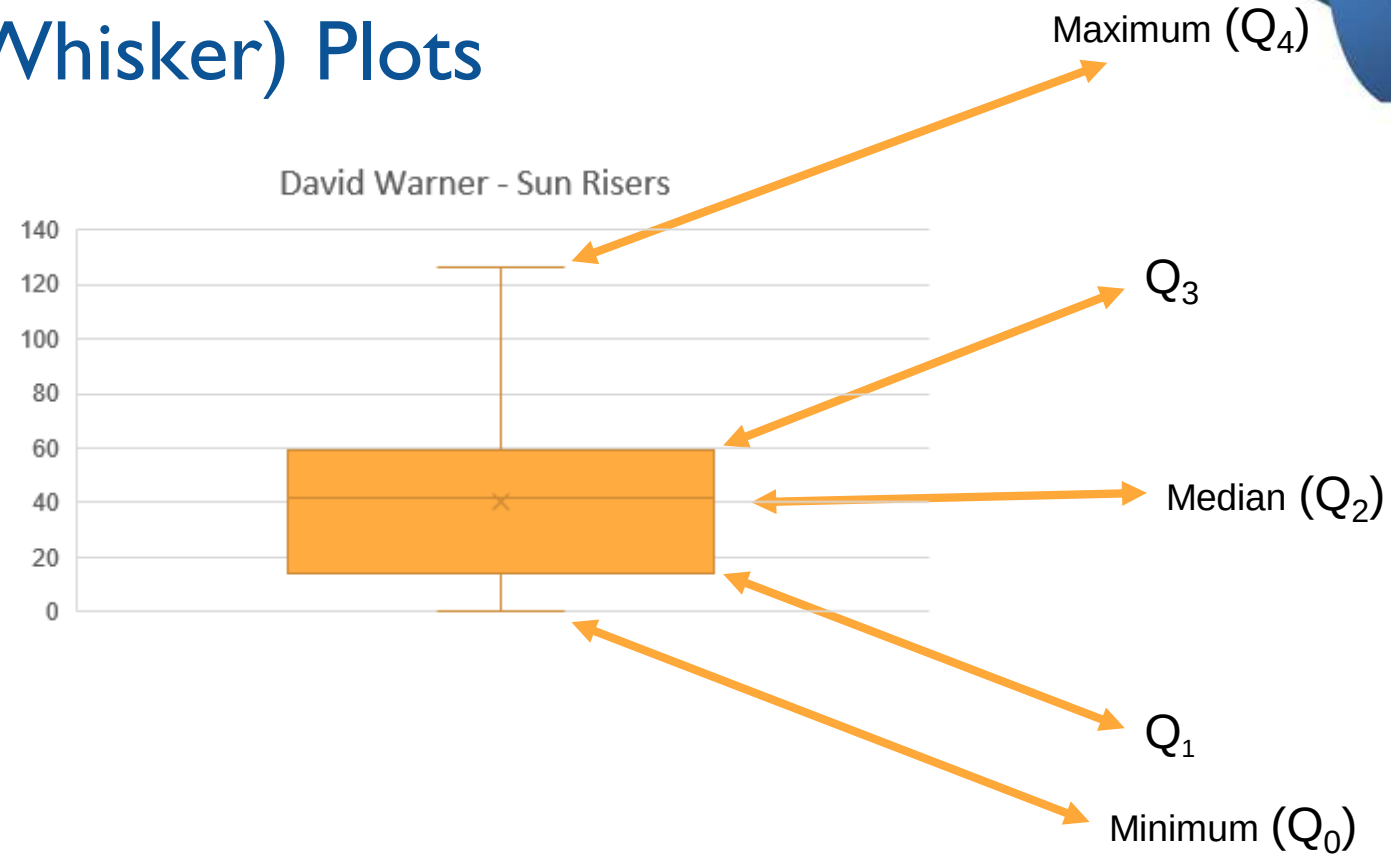
59 - 14 = 45 Runs

Advantages: Good sampling stability, Less influenced by extreme values, appropriate for skewed distributions

Disadvantages: Does not use all the data

Summarizing Quantitative Data – Dispersion – Box (&Whisker) Plots

Minimum (Q_0)	0
Q_1	14
Median (Q_2)	41.5
Q_3	59
Maximum(Q_4)	126



Any data beyond the whiskers is to be inspected – for identifying outliers

David Warner Scores – Standard Deviation

Standard Deviation “s” (for sample) is defined as and it is an indication of “How much do individual observations vary from the Central measure”

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

0	26	57
0	28	58
4	28	58
4	30	59
4	32	59
6	34	59
6	37	61
6	40	61
8	43	69
9	44	73
11	46	73
13	48	81
14	49	90
17	51	91
18	52	92
24	52	126
24	53	

For the given data we can compute this as **28.78 Runs**

The Smaller the standard deviation, the more consistent is a batsman’s performance

Advantages:

- Like mean, uses all the data values
- Has good sampling stability

Disadvantages:

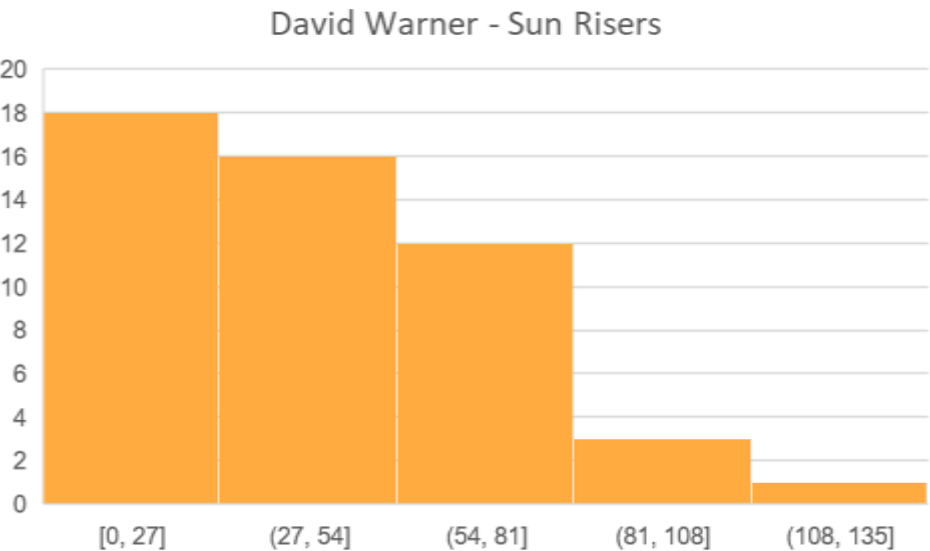
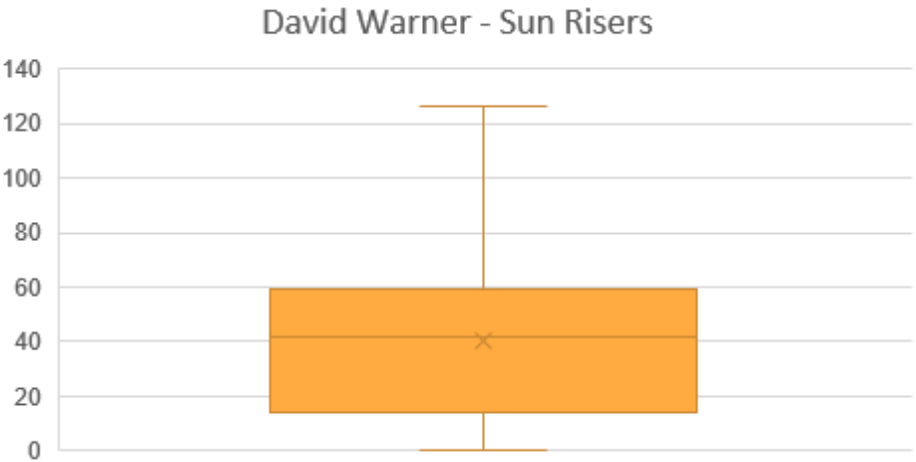
- Is influenced by extreme values/outliers
- Not very appropriate for skewed data

Summary of David Warner's Performance

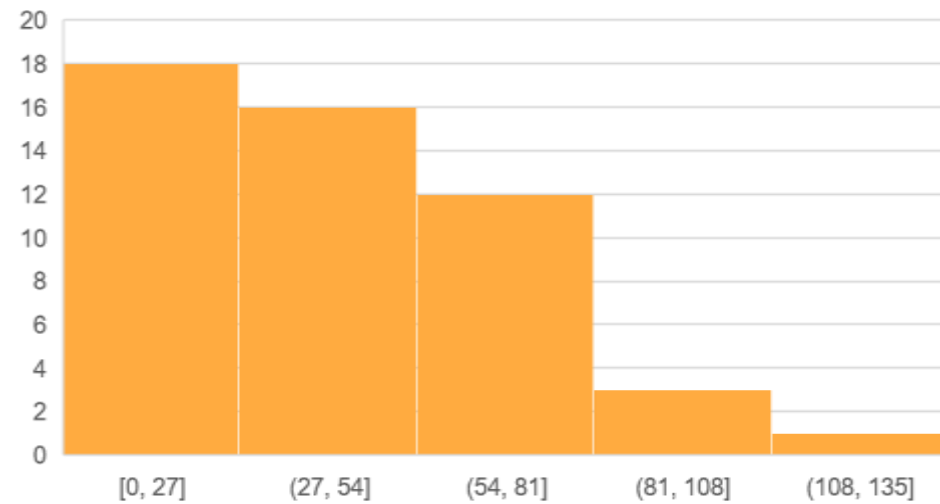
What can we say about Warner's batting performance at Sunrisers Hyderabad?

14	8	126
4	44	37
24	0	51
61	61	0
17	13	73
58	4	24
6	28	73
81	57	59
58	53	6
9	6	48
69	40	4
91	52	43
90	46	18
59	30	28
59	11	49
34	92	26
32	52	

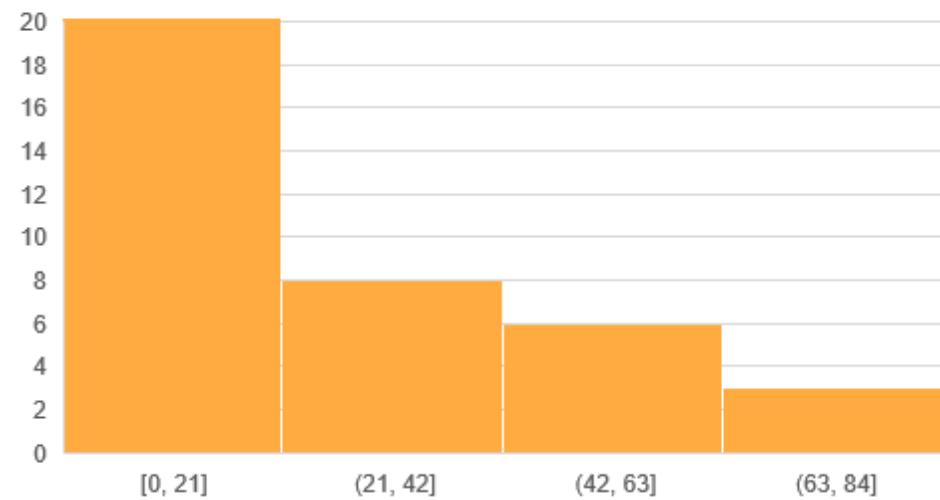
Mean = 40.56 Runs
 Median = 41.50 Runs
 Mode = 4, 6, 59 (6% Each)
 S.D = 28.78 Runs
 IQR = 45 Runs
 Range = 126 Runs



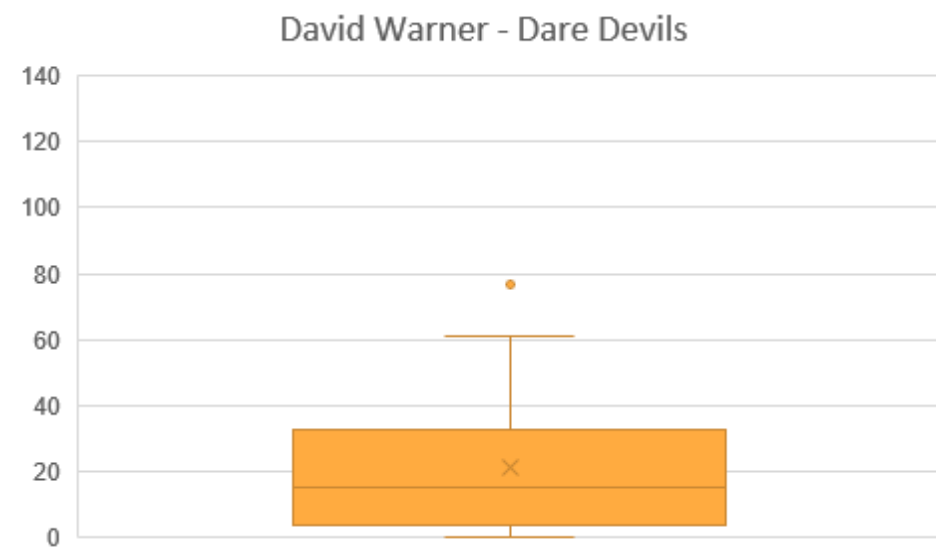
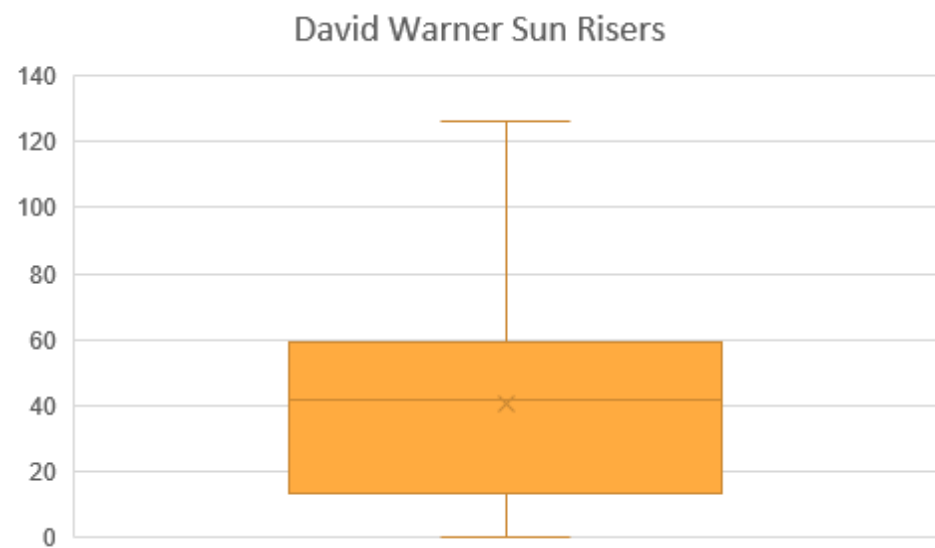
David Warner - Sun Risers



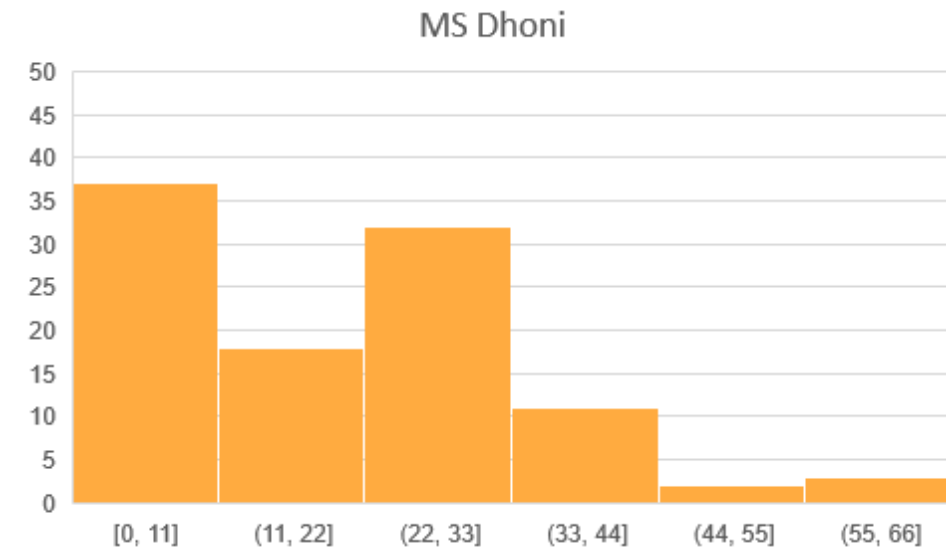
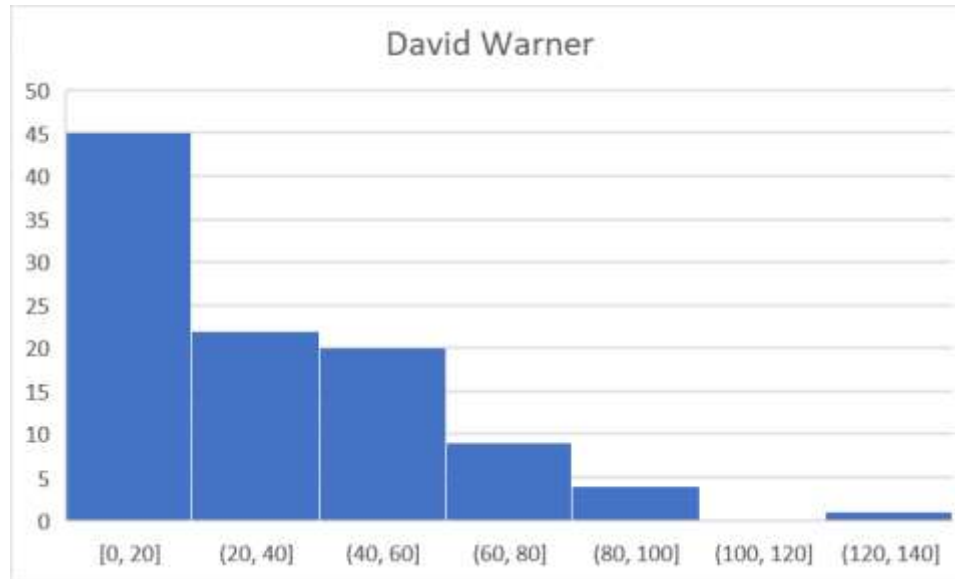
David Warner - Dare Devils



Warner – Hyderabad Vs Delhi

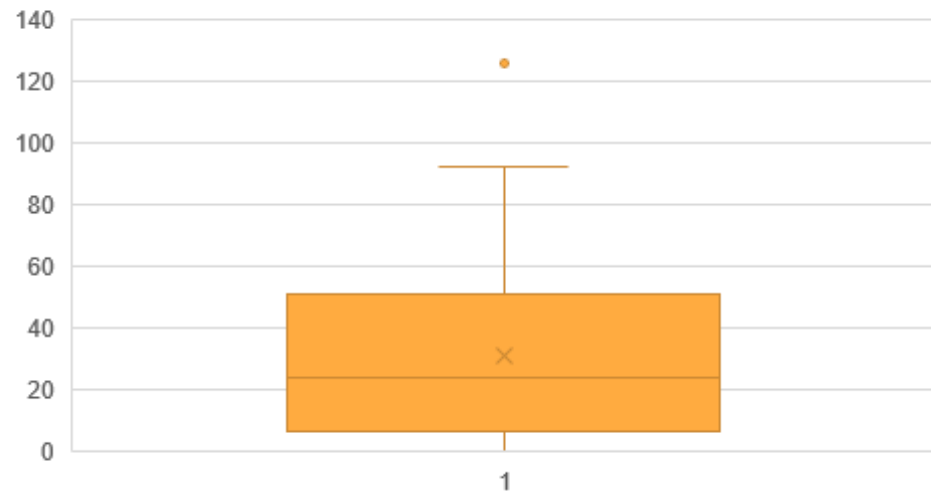


Warner vs Dhoni - Batting



Warner vs Dhoni - Batting

David Warner



MS Dhoni



How Does David Warner get out? (Playing for Sunrisers)

caught

bowled

caught

caught

caught

caught

caught

caught

caught

caught

bowled

run out

caught

caught

caught

caught

run out

lbw

bowled

caught

bowled

caught

lbw

caught

stumped

caught

hit wicket

bowled

bowled

caught

caught

run out

caught

bowled

bowled

caught

caught

caught

run out

caught

lbw

caught

caught

bowled

bowled

bowled

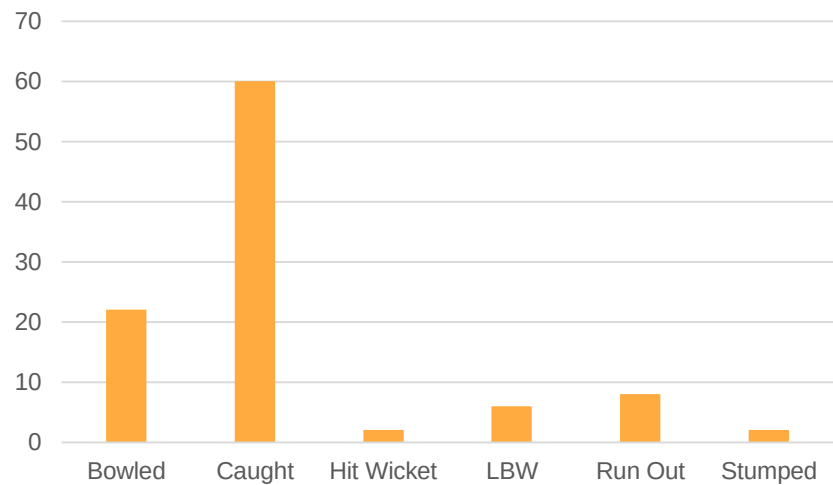
caught

caught

caught

How	# Times	%
Bowled	11	22
Caught	30	60
Hit Wicket	1	2
LBW	3	6
Run Out	4	8
Stumped	1	2

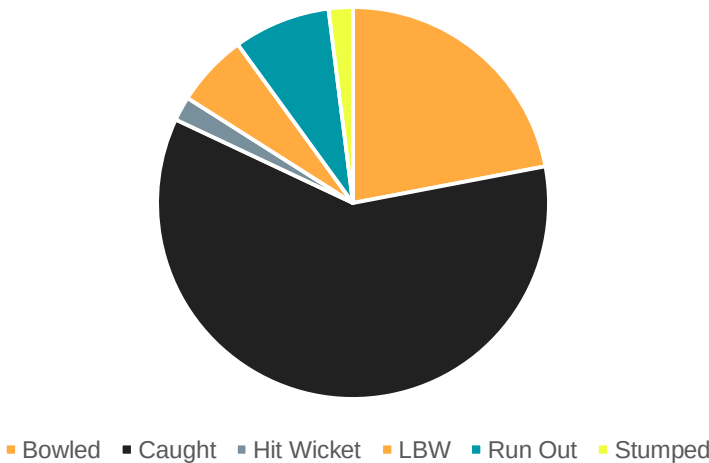
Warner Gets Out %



He gets out most often “Caught”

Mode: Caught (60% of innings)

Warner gets outs out %



How Does David Warner get out? (Playing for Sunrisers)

How can we define variability for a qualitative (Nominal) variable ?

How	# Times	Proportion
Bowled	11	0.22
Caught	30	0.6
Hit Wicket	1	0.02
LBW	3	0.06
Run Out	4	0.08
Stumped	1	0.02

$$\text{Index of qualitative variation} = \frac{1 - \sum_{i=1}^k p_i^2}{(k-1)/k}$$

$$\text{Index of qualitative variation} = \{1 - (0.22)^2 - (0.6)^2 - (0.02)^2 - (0.06)^2 - (0.08)^2 - (0.02)^2\} * 6/5 = \mathbf{0.70}$$

Index of qualitative variation will be **0** if Warner gets out the same way in every match.
That is, **there is no variation in the data**

Index of qualitative variation will be **1** if Warner gets out in each of the ways 16.67% times.
That is, **there is Maximum variation in the data**

If the Index of qualitative variation is close to 0 – No variability

If the Index of qualitative variation is close to 1 – Maximum variability

0.7 can be concluded as reasonably large variation

How Does David Warner get out? (Playing for Sunrisers)

Summary

caught

bowled

caught

caught

caught

caught

caught

caught

caught

caught

caught

bowled

run out

caught

caught

caught

caught

run out

caught

caught

run out

caught

run out

lbw

bowled

caught

caught

bowled

caught

lbw

caught

stumped

caught

hit wicket

bowled

bowled

bowled

caught

caught

run out

caught

bowled

caught

caught

caught

run out

caught

lbw

caught

caught

bowled

bowled

bowled

caught

caught

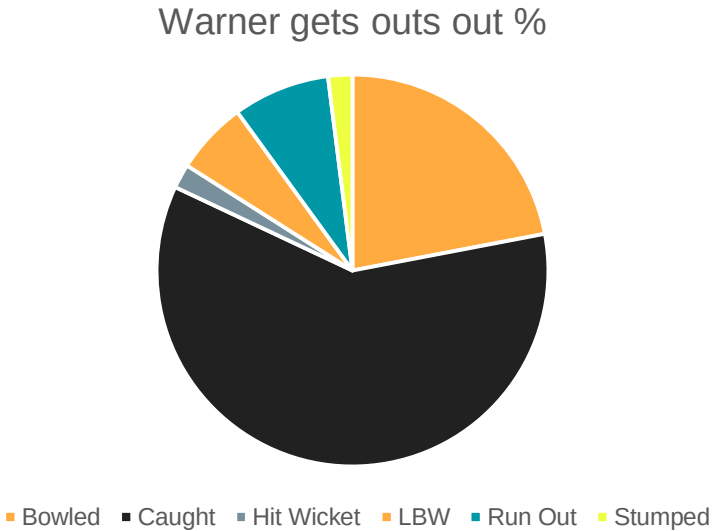
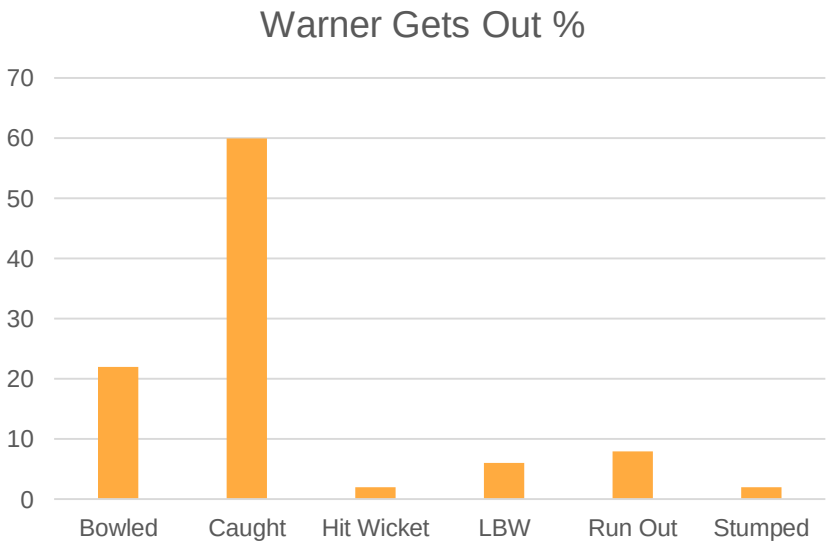
How	# Times	%
Bowled	11	22
Caught	30	60
Hit Wicket	1	2
LBW	3	6
Run Out	4	8
Stumped	1	2

He gets out most often “Caught”

Mode: 60%

Index of Qualitative Variation is 0.7

Reasonably large variation



Association – Two Variables

Does performance of David Warner associated with whether he plays at home or away?

Mean Scores	Hyderabad	Delhi	Total
Home	43	20.43	31.9
Away	38.7	21.9	30
Total	40.5	21.3	30.83

A cross Table is the first step to explore associations between variables

Home or Away does not affect average scores

Franchise does.

Numerical Measure of Associations

Covariance and Correlation

Is the number of wickets a team takes in the first six overs has an association with the outcome of the match?

	Avg # of wickets taken in first 6 Overs	No of matches won
Team A	2.6	8
Team B	1.8	4
Team C	1.3	3
Team D	0.5	3
Average	1.55	4.5

$$COV(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

$COV(Wickets, Wins) =$

$$\frac{(2.6 - 1.55)(8 - 4.5) + (1.8 - 1.55)(4 - 4.5) + (1.3 - 1.55)(3 - 4.5) + (0.5 - 1.55)(3 - 4.5)}{4 - 1}$$

Covariance(Wickets and Wins) = 1.83

Two Issues in interpretation: Covariance is **unbounded**, and the value also **depends on the units** of data

Measure of Associations - Correlations

Correlation between two variables x, y is

$$r = r_{xy} = \frac{\text{Cov}(x, y)}{S_x \times S_y} \quad : \text{Is the } \textbf{Pearson's Correlation} \text{ coefficient}$$

Pearson's correlation coefficient is:

1. Dimensionless, therefore comparable across different variables,
2. Lies between -1, and +1

Correlation

Is the number of wickets a team takes in the first six overs has an association with the outcome of the match?

	Avg # of wickets taken in first 6 Overs	No of matches won
Team A	2.6	8
Team B	1.8	4
Team C	1.3	3
Team D	0.5	3
Average	1.55	4.5
Covariance	1.83	
Correlation	0.87	

Which is more important? Contain or take wickets?

	Avg # of wickets taken in first 6 Overs	No of matches won	Avg # of runs conceded in the first 6 Overs
Team A	2.6	8	36
Team B	1.8	4	42
Team C	1.3	3	38
Team D	0.5	3	40
Average	1.55	4.5	39
Covariance	1.83		-4.00
Correlation	0.87		-0.65

The modulus value of the coefficient indicates the strength of relationship between two variables (Closer it is to 1, the greater the strength)

The Sign of the coefficient indicates the direction of relationship

(+) ---- Variables move in the same direction

(-) ---- Variables move in the opposite direction

```
mirror_object = mirror_object
    operation == "MIRROR_X":
        mirror_mod.use_x = True
        mirror_mod.use_y = False
        mirror_mod.use_z = False
    operation == "MIRROR_Y":
        mirror_mod.use_x = False
        mirror_mod.use_y = True
        mirror_mod.use_z = False
    operation == "MIRROR_Z":
        mirror_mod.use_x = False
        mirror_mod.use_y = False
        mirror_mod.use_z = True
```

```
selection at the end -add
mirror_ob.select= 1
modifier_ob.select=1
context.scene.objects.active
= ("Selected" + str(modifier_ob.name))
mirror_ob.select = 0
= bpy.context.selected_objects[0]
data.objects[one.name].select
print("please select exactly one object")
```

THANK YOU